

C:\Users> Users > Afroz > Desktop > EXP10 > C dataflow.c > ...

```
1  #include <stdio.h>
2  #include <string.h>
3
4  #define MAX 50
5
6  struct Statement
7  {
8      char code[100];
9      int leader;
10 };
11
12 struct Statement stmt[MAX];
13 int n;
14
15 int isJump(char *s)
16 {
17     return (strstr(s, "goto") != NULL ||
18         strstr(s, "if") != NULL);
19 }
20
21 int getTarget(char *s)
22 {
23     char *p;
24     int target;
25
26     p = strstr(s, "goto");
27
28     if (p != NULL)
29     {
30         if (sscanf(p + 4, "%d", &target) == 1)
31             return target;
32     }
33
34     return -1;
35 }
36
37 void findLeaders()
38 {
39     int i, target;
40
41     for (i = 0; i < n; i++)
42         stmt[i].leader = 0;
43
44     stmt[0].leader = 1;
45
46     for (i = 0; i < n; i++)
47     {
48         if (isJump(stmt[i].code))
49         {
50             if (i + 1 < n)
51                 stmt[i + 1].leader = 1;
52
53             target = getTarget(stmt[i].code);
54
55             if (target >= 1 && target <= n)
56                 stmt[target - 1].leader = 1;
57         }
58     }
```

```

58     }
59 }
60
61 void printBasicBlocks()
62 {
63     int i, block = 1;
64
65     printf("\nBASIC BLOCKS\n");
66     printf("=====\n");
67
68     for (i = 0; i < n; i++)
69     {
70         if (stmt[i].leader)
71             printf("\nB%d:\n", block++);
72
73         printf("  %d: %s\n", i + 1, stmt[i].code);
74     }
75 }
76
77 int main()
78 {
79     int i;
80
81     printf("Enter number of three-address statements: ");
82     scanf("%d", &n);
83     getchar();
84
85     printf("\nEnter the statements:\n");
86
87     for (i = 0; i < n; i++)
88     {
89         printf("%d: ", i + 1);
90         fgets(stmt[i].code, sizeof(stmt[i].code), stdin);
91
92         stmt[i].code[strcspn(stmt[i].code, "\n")] = '\0';
93     }
94
95     findLeaders();
96     printBasicBlocks();
97
98     printf("\nData-flow analysis can now be performed "
99         "on these basic blocks.\n");
100
101     return 0;
102 }

```

```
Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/EXP10
$ ./dataflow
Enter number of three-address statements: 7

Enter the statements:
1: a=10
2: b=20
3: c=a+b
4: if c < 50 goto 6
5: c=c+1
6: d=c*2
7: print d

BASIC BLOCKS
=====

B1:
  1: a=10
  2: b=20
  3: c=a+b
  4: if c < 50 goto 6

B2:
  5: c=c+1

B3:
  6: d=c*2
  7: print d

Data-flow analysis can now be performed on these basic blocks.
```